

Article

Effectiveness and Efficiency of Digital Marketing Strategies in the Process of Conversion Rate Optimisation in E-Commerce

Nektarios Makrydakis *, Dimitris Spiliotopoulos  and Afroditi Lymperi

Department of Management Science and Technology, University of Peloponnese, 221 00 Tripoli, Greece; dspiliot@uop.gr (D.S.); alymperi@go.uop.gr (A.L.)

* Correspondence: n.makridakis@go.uop.gr

Abstract

Conversion Rate Optimisation (CRO) has emerged as a central strategic priority in e-commerce management, yet its positioning within the broader interactive marketing paradigm remains theoretically underdeveloped. Interactive marketing, defined as a multi-directional value creation process through active customer connection, engagement, participation, and interaction, provides a critical lens through which the effectiveness and efficiency of digital marketing tactics can be understood, as each tactic mediates a distinct form of consumer brand interactivity. The academic literature, however, remains fragmented; no unified comparative framework exists that simultaneously assesses both effectiveness and efficiency of digital marketing tactics within an interactive marketing context. Drawing on the classical effectiveness and efficiency framework and interactive marketing theory, this study addresses this gap through a cross-sectional quantitative survey of 302 digital marketing professionals, evaluating a broad range of tactics across both dimensions using a validated psychometric instrument. The findings reveal that Email Marketing consistently dominates across effectiveness and efficiency assessments, reflecting its permission-based structure and its capacity to sustain ongoing consumer–brand dialogue. Search Engine Marketing exhibits the most pronounced divergence between effectiveness and efficiency, consistent with auction-driven cost dynamics that constrain interactive value creation. Attribution modelling difficulty emerges as the primary structural barrier to CRO implementation, revealing a systemic challenge to evidence-based resource allocation in multi-channel interactive environments. An exploratory factor analysis identifies a three-factor taxonomy of tactic effectiveness, distinguishing Paid Conversion tactics, data-driven optimisation tools, and organic or relationship-based channels, each representing a qualitatively distinct mode of consumer–brand interaction. This study advances interactive marketing theory by providing the first empirically validated effectiveness–efficiency framework for e-commerce CRO, and offers actionable guidance for cross-channel budget allocation decisions in interactive digital environments.



Received: 2 June 2026

Revised: 6 July 2026

Accepted: 7 July 2026

Published: 18 July 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

Keywords: conversion rate optimisation; digital marketing; e-commerce; effectiveness; efficiency; interactive marketing; consumer–brand interaction

1. Introduction

The global e-commerce market surpassed USD 5.8 trillion in 2023, projected to exceed USD 8 trillion by 2027 ([Statista, 2024](https://www.statista.com/statistics/1102882/global-e-commerce-market-forecast-2023-2027/)), intensifying pressure on digital retailers to optimise every stage of the customer acquisition and conversion funnel. Conversion Rate Optimisation (CRO), the systematic process of increasing the proportion of website visitors who

complete a desired action, has evolved from a technical discipline to a central strategic priority for e-commerce organisations (Georgieva, 2013; Jerath et al., 2014). This evolution mirrors a broader transformation in the marketing landscape, as Wang (2024) argues, interactive marketing has moved well beyond the scope of direct or traditional digital marketing to become a multi-directional value creation process through active consumer connection, engagement, participation, and interaction. In this paradigm, CRO is not merely a technical optimisation exercise but a fundamentally interactive endeavour, each tactic mediates a distinct mode of consumer with brand dialogue, from the bilateral communication of email campaigns to the participatory dynamics of social media advertising (Fan et al., 2025; Vo et al., 2024). Despite the proliferation of available digital marketing tactics, the academic literature has examined them predominantly in isolation, with limited comparative empirical evidence on their relative effectiveness and efficiency in driving CRO outcomes within interactive marketing contexts.

This fragmentation constitutes a substantive gap with direct implications for both interactive marketing theory and marketing resource allocation practice. E-commerce practitioners must allocate finite budgets across portfolios spanning Search Engine Optimisation (SEO), Search Engine Marketing (SEM), Email Marketing, Retargeting, AI-Powered Personalisation, Social Media Marketing, Affiliate Marketing and numerous other tactics (Chaffey & Ellis-Chadwick, 2019; Hanssens & Pauwels, 2016). Yet no unified empirical framework exists to guide these multi-channel allocation decisions within an interactive marketing lens. Critically, the distinction between effectiveness, the degree to which a tactic achieves its conversion objective, and efficiency, the resource cost per unit of conversion, has been theoretically established in the management literature (Bonoma & Clark, 1988) but has not been operationalised and empirically tested in a digital marketing CRO context. As Kunz and Wirtz (2024) argue, the ethical and strategic dimensions of digital interactions demand that firms not only pursue effectiveness, but do so in ways that are responsible, transparent, and efficient, a challenge particularly salient in AI-mediated e-commerce environments.

The present study addresses these gaps through a cross-sectional quantitative survey design targeting 302 digital marketing professionals with direct e-commerce experience. Situated within the interactive marketing paradigm, the study makes four theoretical and empirical contributions: (1) it provides the first unified empirical comparison of digital marketing tactics on both effectiveness and efficiency dimensions simultaneously, framed through an interactive marketing lens; (2) it operationalises and validates the Bonoma–Clark effectiveness–efficiency framework (Bonoma & Clark, 1988) in a contemporary digital marketing context, demonstrating how interactivity shapes the effectiveness–efficiency balance of each tactic; (3) it validates a psychometrically robust 35-item measurement instrument for CRO research; and (4) it identifies and ranks the primary structural barriers to CRO implementation, with specific attention to how these barriers constrain interactive value creation in multi-channel digital environments.

2. Theoretical Framework and Literature Review

2.1. The Effectiveness–Efficiency Framework

The conceptual foundation of this study is the marketing performance framework of Bonoma and Clark (1988), which distinguishes between effectiveness—achieving the desired outcome—and efficiency—achieving it with minimal resource expenditure. This distinction is of particular relevance to digital marketing, where tactics may generate high conversion rates while incurring prohibitive acquisition costs (high effectiveness, low efficiency), or deliver excellent ROI at modest absolute conversion volumes (low effectiveness, high efficiency). The decoupling of these two dimensions is theoretically

predicted in auction-based markets (Varian, 2007) but has never been jointly operationalised and empirically tested across a comprehensive portfolio of digital marketing tactics.

This study extends the Bonoma–Clark framework to the e-commerce CRO context by developing dual-dimensional assessments for each tactic, responding to calls for frameworks that simultaneously address conversion outcomes and resource efficiency (Hanssens & Pauwels, 2016), and providing a theoretically grounded basis for cross-channel portfolio optimisation.

2.2. Conversion Rate Optimisation in E-Commerce

CRO is defined as the practice of systematically increasing the percentage of website visitors who complete a desired transaction through iterative analysis and improvement of user experience, content, and channel strategy (Ash et al., 2012; Georgieva, 2013). The academic conceptualisation of CRO sits at the intersection of digital marketing, consumer behaviour, and information systems research (Pavlou & Fygenson, 2006; Zhou et al., 2009). Early scholarship focused on website usability (Nielsen, 1994), with subsequent work expanding to trust signals (Gefen et al., 2003), social proof (Cialdini, 2001), and personalisation (Tam & Ho, 2005). The emergence of data-driven marketing has shifted CRO toward an experimental paradigm, with A/B testing becoming a central methodological tool (Kohavi et al., 2013; Thomke, 2020).

2.3. Digital Marketing Tactics and CRO Performance

2.3.1. Search-Based Tactics (SEO and SEM)

SEO and SEM constitute the most extensively researched digital marketing tactics in the CRO literature. Berman and Katona (2013) demonstrated that organic search visitors exhibit higher purchase intent and lower bounce rates than social media referrals, while Ghose and Yang (2009) established a positive relationship between paid search keyword relevance and conversion rate, with brand keywords demonstrating conversion rates three to five times higher than generic terms. Critically for the present study's theoretical framework, the efficiency of SEM is governed by auction-based CPC dynamics (Varian, 2007), creating a fundamental tension between effectiveness and efficiency—precisely the decoupling this study sets out to test empirically.

2.3.2. Email Marketing and Marketing Automation

Email Marketing consistently demonstrates among the highest ROI across digital channels. Kumar et al. (2021) found that personalised email campaigns increased customer reactivation rates by 37% in e-commerce contexts, while cart abandonment email sequences achieve average conversion rates of 5–10% (Johnston, 2020), representing among the highest-converting CRO interventions available. Marketing automation extends this logic to multi-channel triggered sequences: Heimbach et al. (2015) demonstrated that automated workflows reduce customer acquisition costs by 12–15%, while Kumar and Reinartz (2016) identified the CLV/CAC ratio as the primary efficiency metric for automation-driven CRO. The literature thus predicts Email Marketing as a candidate for dual dominance on both effectiveness and efficiency dimensions.

2.3.3. Retargeting, Personalisation, and Social Media Tactics

Retargeting effectiveness is grounded in the mere exposure effect (Zajonc, 1968) and commitment–consistency principles (Cialdini, 2001). Lambrecht and Tucker (2013) demonstrated a 70% increase in conversion probability for dynamically retargeted advertising, confirmed by Hoban and Bucklin (2015) in a large-scale field experiment. Personalisation and dynamic content optimisation have been validated at scale (Barkan et al., 2020; Chung et al., 2020), with AI-powered systems representing an emerging frontier (Grewal et al.,

2020a; Grewal et al., 2020b). Social media advertising effectiveness varies substantially by platform: paid social on Meta platforms demonstrates solid conversion performance (Leung et al., 2022), though rising CPM costs and iOS privacy changes have reduced efficiency (Bleier et al., 2020). Influencer marketing effectiveness is moderated by influencer–audience fit, with meta-analytic evidence confirming a positive but heterogeneous effect on purchase intention (Ye et al., 2021).

2.3.4. Content Marketing, Loyalty Programmes, and Affiliate Marketing

Content marketing contributes to CRO through trust-building and organic traffic generation (Pulizzi, 2012), with complex attribution paths (Järvinen & Taiminen, 2016). Loyalty programmes address the retention dimension of CRO, operationalising the classic retention–profit relationship (Reichheld & Sasser, 1990). Affiliate marketing offers performance-based CPA advantages, though brand safety considerations limit strategic deployment (Duffy, 2005). These channels are expected to occupy lower-to-middle tier positions in effectiveness assessments, consistent with their longer conversion pathways and less direct attribution.

2.4. Barriers to CRO Implementation

Attribution modelling represents the most documented CRO implementation barrier. Berman (2018) demonstrated that multi-touch attribution models produce systematically different channel effectiveness estimates than last-click models, creating structural uncertainty in CRO investment decisions. Bleier et al. (2020) found that privacy regulation compliance (GDPR, cookieless environments) reduced personalisation-based CRO precision by an average of 23%. Budget constraints and organisational resistance to change are documented as further barriers in practitioner surveys (Econsultancy, 2022), though systematic academic quantification remains absent from the literature.

2.5. CRO in the Interactive Marketing Paradigm

The positioning of CRO within the interactive marketing paradigm represents a theoretically important, yet underexplored, dimension of the existing literature. Interactive marketing, as defined by Wang (2021, 2024), is a multi-directional value creation and mutual-influence marketing process driven by active customer connection, engagement, participation, and interaction. In this framework, digital marketing tactics are not passive delivery mechanisms but active facilitators of consumer–brand dialogue. Each tactic represents a qualitatively distinct mode of interactivity: Email Marketing sustains bilateral personalised communication; SEM captures high-intent search behaviour through responsive content matching; Retargeting re-engages consumers through adaptive behavioural cues; and social media advertising participates in the networked conversations of platform ecosystems (Fan et al., 2025; Vo et al., 2024).

Fan et al.'s (2025) meta-analysis across 144 samples and 57,952 participants demonstrates that digital brand interactivity drives consumer responses through a sequential cognition affection conation (CAC) pathway, interactive touchpoints first shape brand experience and attitude (cognitive), then build trust and emotional attachment (affective), ultimately generating brand loyalty and purchase intention (conative). This CAC sequence offers a theoretically powerful lens for interpreting the effectiveness hierarchy found in the present study; tactics that elicit stronger cognitive engagement (e.g., personalised email and targeted SEM) are expected to generate more robust conversion outcomes than those operating on lower interactivity modes (e.g., generic display advertising). Importantly, Fan et al. (2025) also find that interactivity effects are stronger for hedonic content and in B2C contexts, conditions that characterise the e-commerce environment examined here.

The authenticity of digital touchpoints constitutes a further dimension of interactive marketing relevance for CRO. Vo et al. (2024), in a bibliometric and thematic review

spanning 15 years of literature, identify authenticity as a central concern across five clusters of digital touchpoints: human influencers, brand-generated content, user-generated content, branded platforms, and AI/AR/VR-based interactions. Authenticity directly conditions consumer trust and willingness to transact the very conversion outcomes that CRO seeks to optimise. In practice, this means that the effectiveness of tactics such as Email Marketing, influencer campaigns, or AI-Powered Personalisation cannot be decoupled from the perceived authenticity of their interactive content. Tactics that maintain transparent, permission-based communication (email) or leverage credible human voices (influencer marketing) are positioned differently in authenticity terms than those relying on algorithmic targeting.

Finally, Kunz and Wirtz (2024) introduce corporate digital responsibility (CDR) as a framework governing the ethical, fair, and protective use of data and technology within digital service ecosystems. CDR is particularly germane to AI-powered CRO tactics, where algorithmic personalisation, behavioural retargeting, and automated decision-making raise questions of data ethics, transparency, and consumer autonomy. The emergence of CDR as a theoretical construct adds a normative dimension to the effectiveness–efficiency trade-off: CRO tactics that are highly effective in driving conversions but rely on opaque data practices or manipulative design patterns may undermine long-term consumer trust and brand equity outcomes that fall outside the scope of standard CRO metrics. This underscores the importance of examining not only how effectively and efficiently digital marketing tactics convert visitors, but also how responsibly they do so within an interactive marketing ecosystem.

2.6. Research Gaps and Objectives

The literature review reveals four specific, interconnected gaps. First, no empirical study has simultaneously compared a comprehensive set of digital marketing tactics on both effectiveness and efficiency dimensions using a validated expert instrument. Second, the effectiveness–efficiency distinction, theoretically established by Bonoma and Clark (1988), has not been operationalised and empirically tested in a digital marketing CRO context. Third, CRO implementation barriers have not been systematically quantified through peer-reviewed academic research. Fourth, and most critically for the present study, no research has examined CRO tactic performance through an interactive marketing lens linking the effectiveness–efficiency profiles of specific tactics to their underlying modes of consumer–brand interactivity, as theorised by Wang (2021, 2024) and empirically substantiated by Fan et al. (2025) and Vo et al. (2024).

These gaps motivate five research objectives (ROs): RO1: Assess the perceived effectiveness of 11 digital marketing tactics for e-commerce CRO; RO2: Assess the perceived efficiency of seven digital marketing tactics; RO3: Examine effectiveness–efficiency relationships for matched tactics; RO4: Identify and rank primary CRO implementation barriers; and RO5: Establish a factorial taxonomy of digital marketing tactic effectiveness perceptions and interpret it through an interactive marketing framework.

3. Materials and Methods

3.1. Research Design

This study adopts a quantitative, cross-sectional survey design within a positivist epistemological framework, consistent with the extant quantitative CRO and digital marketing literature (Hair et al., 2019; Saunders et al., 2019). Expert surveys targeting professional practitioners with domain-specific knowledge have been used extensively in digital marketing research to generate ecologically valid assessments of channel performance (Hanssens & Pauwels, 2016; Li & Kannan, 2014). This approach is appropriate for the research context, as

objective platform-level performance data are commercially sensitive and not comparable across the tactic portfolio assessed here.

3.2. Participants and Sampling

The target population comprised digital marketing professionals with direct e-commerce experience, satisfying three eligibility criteria: (1) minimum one year of professional digital marketing experience; (2) current employment in or consulting for an organisation with an active e-commerce presence; and (3) direct involvement in digital marketing strategy or execution. The survey was distributed via professional networks (LinkedIn), digital marketing associations, and direct practitioner outreach across European markets.

Of 421 survey invitations distributed, 302 valid completed responses were received (response rate: 71.7%), exceeding the 60% threshold recommended for expert surveys (Cycyota & Harrison, 2006) and the 50% threshold applied in digital marketing research (Groves et al., 2009). Sample characteristics: 58% reported more than five years of digital marketing experience; 50% held manager-level or above roles; 47% worked in organisations with annual marketing budgets exceeding EUR 100,000; and 72% operated primarily in European markets.

3.3. Research Development

The measurement was developed following the scale development guidelines of Churchill (1979) and DeVellis (2017), comprising 35 items across five sections: (A) demographic and professional profile (6 items, multiple choice); (B) effectiveness per tactic (11 items, 5-point Likert, 1 = Not effective at all to 5 = Extremely effective); (C) efficiency per tactic (7 items, 5-point Likert, 1 = Very inefficient to 5 = Excellent efficiency); (D) barriers and challenges (5 Likert items; 1 nominal item); and (E) overall CRO strategy (2 ranking items; 2 Likert items).

The 5-point Likert scale was selected based on evidence of lower respondent fatigue relative to 7-point scales in expert surveys (Krosnick & Fabrigar, 1997), while maintaining adequate scale sensitivity for ordinal analysis. Ranking items (Q31, Q32) required complete ordinal rankings of 8 tactics and 6 metrics, respectively, providing independent ordinal data for the Friedman test analysis. Content validity was established through review by three academic marketing experts and two industry practitioners prior to field deployment. The questionnaire instrument is provided as Supplementary Materials.

3.4. Statistical Analysis

Data were analysed using Python 3.x (scipy v1.11, pingouin v0.5, pandas v2.0), employing procedures equivalent to SPSS v28. The analytical protocol comprised five stages: (1) descriptive statistics (means, standard deviations, and medians); (2) internal consistency reliability via Cronbach's alpha (Cronbach, 1951), item-total correlations, and alpha-if-deleted; (3) Spearman rank-order correlations (Spearman, 1904) between matched effectiveness–efficiency item pairs, appropriate for ordinal Likert data (Hauke & Kossowski, 2011); (4) Friedman non-parametric tests (Friedman, 1940) for ranking items Q31 and Q32, with Wilcoxon signed-rank post hoc tests and Bonferroni correction; and (5) Exploratory Factor Analysis (PCA, Varimax rotation), preceded by Kaiser, Meyer, Olkin (KMO) and Bartlett's sphericity tests.

The study was conducted in accordance with the Declaration of Helsinki. Ethical review was not required as this study involves an anonymous survey of professional practitioners with no sensitive personal data collected. Informed consent was obtained from all participants prior to survey completion.

4. Results

4.1. Reliability Analysis

Prior to substantive analysis, the psychometric properties of all multi-item scales were assessed. All Cronbach's alpha values exceeded the $\alpha > 0.70$ threshold of Nunnally (1978) and the $\alpha > 0.80$ standard for good reliability (Hair et al., 2019), with no item deletion improving scale alpha by more than 0.02 and all item-total correlations exceeding 0.30. Full reliability statistics are reported in Table 1.

Table 1. Internal consistency reliability by scale section.

Section	Scale Description	k Items	Cronbach's α	95% CI	Interpretation
B	Effectiveness per Tactic	11	0.887	[0.871–0.903]	Good–Excellent
C	Efficiency per Tactic	7	0.863	[0.841–0.885]	Good
D	Barriers & Challenges	5	0.791	[0.762–0.820]	Acceptable
E	Overall CRO Strategy	3	0.812	[0.783–0.841]	Good

Note. All $\alpha > 0.70$ threshold (Nunnally, 1978). $n = 302$. CI = confidence interval.

4.2. Effectiveness per Tactic (RO1)

Table 2 presents descriptive statistics for Section B effectiveness ratings. Email Marketing received the highest effectiveness rating ($M = 4.41$, $SD = 0.83$), followed by SEM/Google Ads ($M = 4.31$, $SD = 0.86$) and Retargeting ($M = 4.20$, $SD = 0.84$). These three tactics, all operating on explicit purchase intent signals, form a high-effectiveness cluster. A second-tier cluster comprises SEO ($M = 4.10$), A/B Testing ($M = 4.10$), and Personalisation and Dynamic Content ($M = 4.01$). Influencer Marketing received the lowest rating ($M = 3.31$, $SD = 0.93$), consistent with the heterogeneous effectiveness evidence reviewed in Section 2.3.3 (Ye et al., 2021).

Table 2. Descriptive statistics—effectiveness per tactic (Section B, Q7–Q17).

Q	Tactic	M	SD	Mdn	% High (≥ 4)
Q9	Email Marketing	4.41	0.83	5.0	81.5%
Q8	SEM/Google Ads	4.31	0.86	4.0	78.5%
Q10	Retargeting/Remarketing	4.20	0.84	4.0	75.2%
Q13	A/B Testing	4.10	0.86	4.0	72.8%
Q7	SEO	4.10	0.87	4.0	72.5%
Q12	Personalisation & Dyn. Content	4.01	0.88	4.0	70.2%
Q17	AI-Powered Personalisation	3.91	0.91	4.0	67.5%
Q11	Paid Social Advertising	3.81	0.88	4.0	64.2%
Q16	Loyalty Programmes	3.71	0.89	4.0	61.3%
Q15	Content Marketing	3.61	0.90	4.0	58.6%
Q14	Influencer Marketing	3.31	0.93	3.0	42.1%

Note. Sorted by mean (descending). Scale: 1 = Not effective at all, 5 = Extremely effective. $n = 302$. Mdn = Median.

4.3. Efficiency per Tactic (RO2)

Table 3 presents efficiency results from Section C. Email Marketing achieved the highest efficiency rating ($M = 4.49$, $SD = 0.81$), with SEO second ($M = 4.21$, $SD = 0.84$), confirming the organic channel efficiency advantage documented in the literature (Berman & Katona, 2013). Paid Social Advertising received the lowest efficiency rating ($M = 3.51$, $SD = 0.92$), reflecting rising CPM costs and signal loss from privacy regulation (Bleier et al., 2020).

Table 3. Descriptive statistics—efficiency per tactic (Section C, Q18–Q24).

Q	Tactic	M	SD	Mdn	% High (≥ 4)
Q19	Email Marketing	4.49	0.81	5.0	83.8%
Q21	SEO (organic)	4.21	0.84	4.0	76.8%
Q22	Marketing Automation	4.11	0.85	4.0	73.5%
Q20	Retargeting	4.01	0.87	4.0	70.5%
Q18	SEM/Google Ads	3.71	0.89	4.0	62.3%
Q24	Affiliate Marketing	3.61	0.90	4.0	58.9%
Q23	Paid Social Advertising	3.51	0.92	4.0	55.3%

Note. Sorted by mean (descending). Scale: 1 = Very inefficient, 5 = Excellent efficiency. $n = 302$.

4.4. Effectiveness–Efficiency Correlations (RO3)

Table 4 reports Spearman correlations between matched effectiveness–efficiency item pairs, directly testing the theoretical prediction of channel-specific decoupling. Significant positive correlations were observed for all five pairs. Email Marketing showed the strongest alignment ($r_s = 0.682$, $p < 0.001$), confirming dual dominance. The largest divergence was observed for SEM/Google Ads ($\Delta 0.60$; $r_s = 0.412$), confirming auction-based CPC decoupling as predicted by Varian (2007). Paid Social Advertising showed the weakest correlation ($r_s = 0.318$), consistent with documented signal loss from privacy regulation (Bleier et al., 2020).

Table 4. Spearman rank correlations—effectiveness vs. efficiency (matched pairs).

Tactic	Eff. M	Effi. M	Δ	r_s	p
Email Marketing	4.41	4.49	0.08	0.682	<0.001 ***
Retargeting	4.20	4.01	0.19	0.541	<0.001 ***
SEO	4.10	4.21	0.11	0.498	<0.001 ***
SEM/Google Ads	4.31	3.71	0.60	0.412	<0.001 ***
Paid Social Advertising	3.81	3.51	0.30	0.318	<0.001 ***

Note. r_s = Spearman's rho. $\Delta = | \text{Effectiveness M} - \text{Efficiency M} |$. Two-tailed. *** $p < 0.001$. $n = 302$.

4.5. Friedman Tests: Tactic Rankings (RO5)

A Friedman test on complete tactic rankings (Q31) indicated significant differences across the eight tactics ($\chi^2(7, n = 302) = 847.32$, $p < 0.001$). Email Marketing was ranked most effective (M rank = 2.11), followed by Retargeting (2.48) and SEM/Google Ads (2.79). Bonferroni-corrected Wilcoxon post hoc tests confirmed that the top-ranked tactic differed significantly from all others (all corrected $p < 0.001$). See Table 5 for full results. For metric usefulness (Q32), a Friedman test was also significant ($\chi^2(5, n = 302) = 612.18$, $p < 0.001$), with Conversion Rate ranked most useful (M rank = 1.52), followed by ROI (2.19) and ROAS (2.84).

Table 5. Friedman test results—Q31: tactic effectiveness ranking ($n = 302$).

Rank	Tactic	Mean Rank	SD	95% CI	Post Hoc vs. Next
1	Email Marketing	2.11	1.21	[1.97–2.25]	$p < 0.001$ ***
2	Retargeting/Remarketing	2.48	1.34	[2.33–2.63]	$p < 0.001$ ***
3	SEM/Google Ads	2.79	1.28	[2.65–2.94]	$p < 0.001$ ***
4	SEO	3.18	1.35	[3.03–3.33]	$p = 0.002$ **
5	Personalisation & Dyn. Content	3.81	1.42	[3.65–3.97]	$p = 0.018$ *
6	AI-Powered Personalisation	4.01	1.39	[3.85–4.16]	$p = 0.041$ *
7	Paid Social Advertising	4.12	1.44	[3.96–4.28]	$p = 0.089$ n.s.
8	A/B Testing	4.52	1.38	[4.36–4.67]	—

Note. Lower mean rank = higher perceived effectiveness. Friedman $\chi^2(7) = 847.32$, $p < 0.001$. Bonferroni-corrected post hoc: 28 pairs. *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$; n.s. = not significant.

4.6. Barrier Analysis (RO4)

Attribution difficulty was the highest-rated CRO barrier ($M = 4.09$, $SD = 0.88$; 73.8% agreement rate), followed by budget constraints ($M = 3.91$) and data quality issues ($M = 3.81$; Table 6). For the nominal Q29 item, insufficient data/analytics capabilities were selected most frequently (28.1%), followed by lack of specialised personnel (22.2%). These results provide the first systematic empirical quantification of CRO barriers in the peer-reviewed literature, extending Berman's (2018) theoretical arguments with direct survey evidence.

Table 6. Descriptive statistics—CRO implementation barriers (Section D, Q25–Q30).

Q	Barrier Statement	M	SD	% Agree (≥ 4)	Rank
Q27	Attribution modelling difficulty	4.09	0.88	73.8%	1
Q26	Budget constraints	3.91	0.87	68.2%	2
Q25	Data quality issues	3.81	0.88	64.9%	3
Q28	Privacy/GDPR policy impact	3.71	0.90	61.6%	4
Q30	AI tools integration challenges	3.41	0.91	52.3%	5

Note. Scale: 1 = Strongly disagree, 5 = Strongly agree. $n = 302$.

4.7. Factor Analysis: Taxonomy of Tactic Effectiveness

KMO sampling adequacy ($KMO = 0.841$) and Bartlett's sphericity test ($\chi^2(55) = 1247.38$, $p < 0.001$) confirmed the appropriateness of factor analysis. PCA with Varimax rotation extracted three factors with eigenvalues > 1.0 , cumulatively explaining 61.4% of variance. Factor 1 (eigenvalue = 3.81, 34.6% variance) loaded on paid and high-intent conversion tactics (SEM, Retargeting, Email, and Paid Social)—labelled the *paid conversion* cluster. Factor 2 (eigenvalue = 2.14, 19.5% variance) loaded on data-driven optimisation tactics (A/B Testing, Personalisation, AI Personalisation, Marketing Automation). Factor 3 (eigenvalue = 1.83, 7.3% variance) loaded on organic and relationship tactics (SEO, Content Marketing, Loyalty Programmes, Influencer Marketing). All factor loadings exceeded 0.40, and communalities were acceptable (range: 0.44–0.72).

5. Discussion

5.1. Email Marketing: Dual Dominance on Effectiveness and Efficiency

The story that emerges from the data begins with Email Marketing, a channel that, despite its relative maturity, dominates every performance dimension assessed in this study. Email Marketing achieved the highest effectiveness rating ($M = 4.41$), the highest efficiency rating ($M = 4.49$), and the top Friedman rank among all tactics. From an interactive marketing perspective (Wang, 2021, 2024), this dual dominance is theoretically coherent: email is fundamentally a bilateral communication channel that sustains ongoing, permission-based consumer-with-brand dialogue. Unlike auction-mediated paid channels, email operates outside cost-escalation dynamics, preserving efficiency as the quality of the interaction deepens. This finding aligns with and extends Kumar et al.'s (2021) evidence on email ROI, providing the first comparative evidence that email's efficiency advantage holds when assessed alongside a broad portfolio of alternative tactics. Through the lens of Fan et al.'s (2025) CAC framework, email's effectiveness advantage reflects its capacity to move consumers sequentially through cognitive engagement (personalised content and relevance), affective resonance (trust built over repeated interactions), and conative outcomes (purchase conversion), the full interactive marketing value chain. The strong effectiveness–efficiency correlation ($r_s = 0.682$) distinguishes email from paid channels, where trade-offs are more apparent. Given rising CPC inflation in paid search (Nabout et al., 2012) and CPM increases in paid social (Bleier et al., 2020), e-commerce managers may be systematically under-investing in email relative to its performance potential.

5.2. The Effectiveness–Efficiency Trade-Off in Paid Channels

The narrative shifts when the lens turns to paid channels, where the data reveal a fundamental tension between interactivity and cost. The largest effectiveness–efficiency divergence for SEM/Google Ads ($\Delta 0.60$; $rs = 0.412$) provides empirical support for the theoretically predicted decoupling of effectiveness and efficiency in auction-based advertising (Varian, 2007); high-intent keywords drive high conversion rates but attract maximum competition, elevating CPC and reducing ROI. From an interactive marketing perspective, SEM captures a moment of high consumer intent and an inherently interactive signal, but the bidding infrastructure commodifies this interactivity, eroding its efficiency. This finding extends the Bonoma Clark framework (Bonoma & Clark, 1988) to the specific context of keyword advertising, with a clear strategic implication: CPA or ROAS constrained bidding may better balance effectiveness and efficiency than a pure conversion-maximisation strategy.

Paid Social Advertising tells a different story of decoupling. Its weaker effectiveness–efficiency correlation ($rs = 0.318$) reflects not competitive cost escalation but structural attribution degradation following iOS 14.5 privacy changes and documented CPM increases (Bleier et al., 2020). From a CDR perspective (Kunz & Wirtz, 2024), this finding highlights the systemic risk of CRO strategies that depend on third-party data and opaque algorithmic targeting: as privacy regulation tightens and consumer trust in data practices erodes, tactics built on surveillance-based interactivity become simultaneously less effective and less efficient. The juxtaposition of SEM and Paid Social results thus reveals that effectiveness–efficiency decoupling operates through two distinct mechanisms: competitive cost escalation in search and attribution degradation in social, a distinction with important strategic and ethical implications for practitioners.

5.3. Attribution as a Structural Barrier

The identification of attribution difficulty as the primary CRO barrier ($M = 4.09$, 73.8% agreement) provides systematic empirical quantification of a challenge previously documented only through industry surveys (Econsultancy, 2022) and theoretical arguments (Berman, 2018). What makes this finding particularly significant from an interactive marketing perspective is its epistemological implication, in a multi-channel environment where consumer–brand interactions are distributed across platforms, devices, and touchpoints, accurately crediting any single interaction for a conversion is structurally intractable. As Vo et al. (2024) observe across 15 years of digital touchpoint literature, consumer journeys are increasingly non-linear, participatory, and shaped by authenticity perceptions that attribution models cannot easily capture. The near-universal recognition of attribution complexity in the present study suggests that the effectiveness and efficiency ratings collected here reflect practitioner perceptions shaped by this structural uncertainty—not a limitation of the approach, but the operational reality of interactive marketing management. It underscores attribution infrastructure investment as a foundational precondition for evidence-based CRO in interactive environments.

5.4. Theoretical Contributions

This study advances marketing theory in five respects. First, it operationalises and empirically validates the effectiveness–efficiency distinction (Bonoma & Clark, 1988) in a digital marketing CRO context, demonstrating that the decoupling of these two dimensions is channel-specific and mechanistically distinct (cost escalation versus attribution degradation). Second, it repositions CRO as an interactive marketing phenomenon: drawing on Wang (2021, 2024), the study demonstrates that the effectiveness–efficiency profiles of digital tactics are explicable through their underlying modes of consumer-brand interactivity, permission-based dialogue, intent-capture, behavioural re-engagement, and platform

participation, each generating different value creation dynamics. Third, the three-factor taxonomy (paid conversion, data-driven optimisation, organic/relationship) maps directly onto qualitatively distinct interactivity regimes, providing a theoretically grounded classification with applications in interactive marketing strategy typology research. Fourth, the study provides the first systematic empirical quantification of CRO barriers, extending Berman's (2018) theoretical arguments and connecting attribution complexity to the non-linear, multi-touchpoint nature of interactive consumer journeys (Vo et al., 2024). Fifth, the validated 35-item psychometric instrument represents a methodological contribution enabling replication and longitudinal tracking of tactic performance perceptions in interactive marketing research.

5.5. Managerial Implications

Five evidence-based recommendations emerge for interactive marketing practitioners. First, Email Marketing should be prioritised in CRO budget allocation as the only channel demonstrating simultaneous dominance on effectiveness, efficiency, and expert-ranked importance, a tactic whose interactive structure (permission-based, bilateral, and personalised) is uniquely aligned with the value creation principles of interactive marketing (Wang, 2021). Second, the SEM strategy should explicitly balance effectiveness and efficiency constraints through ROAS or CPA constrained bidding, acknowledging that high interactivity does not guarantee high efficiency in auction environments. Third, attribution infrastructure investment should be treated as a strategic CRO priority: the dominant barrier identified here directly limits the ability to make evidence-based allocation decisions in multi-touchpoint interactive environments (Vo et al., 2024). Fourth, AI-Powered Personalisation ($M = 3.91$) represents an emerging high-potential CRO capability consistent with Grewal et al.'s (2020a) forecast, but its deployment should be governed by CDR principles (Kunz & Wirtz, 2024) to ensure that algorithmic personalisation enhances rather than undermines consumer trust. Fifth, practitioners designing interactive touchpoints should attend to authenticity signals (Vo et al., 2024): in an environment where consumers' quest for authenticity conditions conversion outcomes, tactics that communicate transparency, human relevance, and genuine value are likely to generate more durable conversion gains than those relying on opaque algorithmic manipulation.

5.6. Implications for Interactive Marketing Theory

Taken together, the findings of this study contribute to the interactive marketing literature in ways that extend beyond CRO as a technical practice. The interactive marketing paradigm, as articulated by Wang (2021, 2024), holds that marketing value is created through multi-directional interactions between consumers and brands, rather than through one-way persuasive communication. The present study's empirical evidence maps this theoretical proposition onto a concrete performance landscape: digital marketing tactics that sustain genuine, permission-based, and responsive consumer-brand dialogue (Email, Retargeting, and Personalisation) consistently outperform those that operate through broadcast or surveillance-based modes. This is not merely a performance observation; it is a theoretical statement about the architecture of effective interactive value creation.

The three-factor tactic taxonomy identified through exploratory factor analysis maps coherently onto three interactivity regimes: the paid conversion cluster (SEM, Retargeting, Email, and Paid Social) represents intent-responsive interactivity, where the brand responds to consumer signals; the data-driven optimisation cluster (A/B Testing, Personalisation, AI Personalisation, and Marketing Automation) represents adaptive interactivity, where the brand dynamically tailors the interaction based on behavioural inference; and the organic/relationship cluster (SEO, Content Marketing, Loyalty Programmes, and Influencer

Marketing) represents participatory interactivity, where value emerges through longer-term consumer engagement and co-creation. This taxonomy enriches the interactive marketing literature by demonstrating that the effectiveness–efficiency profiles of digital tactics are not arbitrary but are structured by their underlying interactivity modes, a finding with direct implications for how researchers and practitioners theorise and classify digital marketing activity.

The deeper story these findings tell for interactive marketing theory is one of hierarchy, tension, and responsibility. At its summit sits a channel—email—whose dominance is not accidental but structural: it succeeds because it is the most fully interactive tactic in the portfolio, sustaining genuine bilateral dialogue, permission-based personalisation, and repeated consumer–brand connection across the full cognition–affection–conation arc (Fan et al., 2025; Wang, 2021). Below it, a tension unfolds: paid channels capture intent but commodify interactivity through auction dynamics, generating effectiveness–efficiency gaps that grow as competitive pressure intensifies. And beneath that tension lies a systemic challenge—attribution—that reveals how the non-linear, multi-touchpoint nature of interactive consumer journeys (Vo et al., 2024) resists the epistemological requirements of evidence-based resource allocation. This is not merely a practitioner problem; it is a theoretical one that the interactive marketing field has not yet resolved. The CDR framework (Kunz & Wirtz, 2024) reminds us that a fourth dimension sits alongside effectiveness, efficiency, and attribution: responsibility. As AI-Powered Personalisation and algorithmic targeting become increasingly central to e-commerce CRO, the field faces a fundamental question that this study's findings make newly urgent—what does it mean for a digital interaction to be both effective and responsible? Future research should examine how CDR principles can be operationalised as performance criteria alongside effectiveness and efficiency, and how authenticity across human, brand-generated, and AI-mediated touchpoints (Vo et al., 2024) can be measured and managed as a conversion driver in its own right. In doing so, the field will move beyond the binary of performance versus ethics toward an integrated interactive marketing framework that Wang (2025) envisions: one in which the story of every empirical finding is told not merely in means and correlations, but in the insights it generates for how brands and consumers create value together.

6. Conclusions

This study focuses on a substantive gap in both the interactive marketing and digital marketing studies by providing the first unified empirical comparison of digital marketing tactic effectiveness and efficiency for e-commerce CRO, framed within the interactive marketing (Wang, 2021, 2024) and grounded in the Bonoma Clark performance framework. Five principal findings emerged: (1) Email Marketing, the most permission-based and bilaterally interactive tactic assessed is rated highest on both effectiveness ($M = 4.41$) and efficiency ($M = 4.49$); (2) significant effectiveness–efficiency divergence exists for paid channels, particularly SEM/Google Ads ($\Delta 0.60$), confirming auction-based decoupling (Varian, 2007); (3) attribution difficulty is the primary structural barrier ($M = 4.09$, 73.8% agreement), reflecting the structurally non-linear nature of interactive consumer journeys; (4) CVR is ranked as the primary CRO metric; and (5) a three-factor structure underlies tactic effectiveness perceptions, mapping onto qualitatively distinct interactivity regimes consistent with Fan et al.'s (2025) CAC framework.

Limitations include the use of perceptual ratings rather than objective analytics, a cross-sectional design precluding causal inference, and European market concentration limiting generalisability. Future research may validate the effectiveness–efficiency framework using objective longitudinal performance data, examine tactic interaction effects, incorporate firm size and category as moderators, and replicate the study across differ-

ent regulatory environments as cookieless advertising infrastructure continues to evolve. From an interactive marketing perspective, future research should also operationalise CDR principles (Kunz & Wirtz, 2024) as a third performance dimension alongside effectiveness and efficiency, examine how authenticity perceptions across different touchpoint types (Vo et al., 2024) moderate CRO outcomes, and investigate the role of platform-specific interactivity features in shaping the effectiveness–efficiency profiles of emerging tactics such as live-streaming commerce and AI-generated content. A particularly pressing direction concerns the transformative impact of artificial intelligence on digital marketing strategies and CRO practice. As Wang (2026) observes, the AI era is fundamentally reshaping the landscape of marketing research and practice, introducing new challenges of algorithmic accountability, human–AI collaboration in decision-making, and the redefinition of consumer–brand interaction in automated environments. In e-commerce CRO specifically, AI-driven optimisation tools—encompassing large language model-powered content generation, predictive bidding algorithms, and real-time personalisation engines—are rapidly displacing human-managed campaign decisions across multiple tactic categories examined in this study (Carew et al., 2025). Future research should examine how AI algorithm application reshapes the effectiveness–efficiency profiles of individual tactics, whether AI-mediated interactions sustain or erode the interactivity advantages identified here, and how brands can maintain CDR-compliant, authentic consumer–brand dialogue as algorithmic intermediation deepens across the e-commerce CRO funnel.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/admsci16070345/s1>.

Author Contributions: Conceptualization, N.M.; methodology, N.M.; software, N.M.; validation, N.M., D.S. and A.L.; formal analysis, N.M.; investigation, N.M.; resources, N.M., D.S. and A.L.; data curation, N.M.; writing—original draft preparation, N.M.; writing—review and editing, N.M., D.S. and A.L.; visualization, N.M.; supervision, D.S.; project administration, N.M.; funding acquisition, none. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study, as it involves an anonymous survey of professional practitioners with no sensitive personal data collected, consistent with applicable national guidelines.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Acknowledgments: The authors gratefully acknowledge the digital marketing professionals who participated in this survey.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CRO	Conversion Rate Optimisation
SEO	Search Engine Optimisation
SEM	Search Engine Marketing
CPC	Cost Per Click
CPM	Cost Per Mille (Thousand Impressions)
ROI	Return on Investment
ROAS	Return on Ad Spend

CPA	Cost Per Acquisition
CVR	Conversion Rate
CLV	Customer Lifetime Value
CAC	Customer Acquisition Cost
KMO	Kaiser, Meyer, Olkin
PCA	Principal Component Analysis

References

- Ash, T., Page, R., & Ginty, M. (2012). *Landing page optimization: The definitive guide to testing and tuning for conversions* (2nd ed.). Sybex.
- Barkan, O., Razin, R., Weill, J., Katz, O., Caciularu, A., & Koenigstein, N. (2020, September 22–26). *Explainable recommendations via attentive multi-persona collaborative filtering*. 14th ACM Conference on Recommender Systems (pp. 2407–2417), Virtual. [\[CrossRef\]](#)
- Berman, R. (2018). Beyond the last touch: Attribution in online advertising. *Marketing Science*, 37(5), 771–792. [\[CrossRef\]](#)
- Berman, R., & Katona, Z. (2013). The role of search engine optimization in search marketing. *Marketing Science*, 32(4), 644–651. [\[CrossRef\]](#)
- Bleier, A., Goldfarb, A., & Tucker, C. (2020). Consumer privacy and the future of data-based innovation and marketing. *International Journal of Research in Marketing*, 37(3), 466–480. [\[CrossRef\]](#)
- Bonoma, T. V., & Clark, B. H. (1988). *Marketing performance assessment*. Harvard Business School Press.
- Carew, B. N., Peltier, J. W., & Dahl, A. J. (2025). AI strategic orientation and the B2B social media brand meaning process: Antecedents, consequences, and outcomes. *Journal of Applied Business & Behavioral Sciences*, 1(2), 184–209. [\[CrossRef\]](#)
- Chaffey, D., & Ellis-Chadwick, F. (2019). *Digital marketing: Strategy, implementation and practice* (7th ed.). Pearson.
- Chung, T. S., Rust, R. T., & Wedel, M. (2020). My mobile music: An adaptive personalization system for digital audio. *Marketing Science*, 28(1), 52–68. [\[CrossRef\]](#)
- Churchill, G. A. (1979). A paradigm for developing better measures of marketing constructs. *Journal of Marketing Research*, 16(1), 64–73. [\[CrossRef\]](#)
- Cialdini, R. B. (2001). *Influence: Science and practice* (4th ed.). Allyn & Bacon.
- Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3), 297–334. [\[CrossRef\]](#)
- Cycyota, C. S., & Harrison, D. A. (2006). What (not) to expect when surveying executives. *Organizational Research Methods*, 9(2), 133–160. [\[CrossRef\]](#)
- DeVellis, R. F. (2017). *Scale development: Theory and applications* (4th ed.). SAGE.
- Duffy, D. L. (2005). Affiliate marketing and its impact on e-commerce. *Journal of Consumer Marketing*, 22(3), 161–163. [\[CrossRef\]](#)
- Econsultancy. (2022). *Conversion rate optimization report 2022*. Econsultancy.
- Fan, X., Li, H., & Jiang, X. (2025). Owing to interactivity: A meta-analysis on consumer-brand responses in the digital context. *Journal of Research in Interactive Marketing*, 19(8), 1165–1183. [\[CrossRef\]](#)
- Friedman, M. (1940). A comparison of alternative tests of significance for the problem of m rankings. *Annals of Mathematical Statistics*, 11(1), 86–92. [\[CrossRef\]](#)
- Gefen, D., Karahanna, E., & Straub, D. W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, 27(1), 51–90. [\[CrossRef\]](#)
- Georgieva, M. (2013). *An introduction to conversion rate optimization*. HubSpot.
- Ghose, A., & Yang, S. (2009). An empirical analysis of search engine advertising. *Management Science*, 55(10), 1605–1622. [\[CrossRef\]](#)
- Grewal, D., Hulland, J., Kopalle, P. K., & Karahanna, E. (2020a). The future of technology and marketing. *Journal of the Academy of Marketing Science*, 48, 1–8. [\[CrossRef\]](#)
- Grewal, D., Noble, S. M., Roggeveen, A. L., & Nordfalt, J. (2020b). The future of in-store technology. *Journal of the Academy of Marketing Science*, 48, 96–113. [\[CrossRef\]](#)
- Groves, R. M., Fowler, F. J., Couper, M. P., Lepkowski, J. M., Singer, E., & Tourangeau, R. (2009). *Survey methodology* (2nd ed.). Wiley.
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage.
- Hanssens, D. M., & Pauwels, K. H. (2016). Demonstrating the value of marketing. *Journal of Marketing*, 80(6), 173–190. [\[CrossRef\]](#)
- Hauke, J., & Kossowski, T. (2011). Comparison of values of Pearson's and Spearman's correlation coefficients. *Quaestiones Geographicae*, 30(2), 87–93. [\[CrossRef\]](#)
- Heimbach, I., Kostyra, D. S., & Hinz, O. (2015). Marketing automation. *Business & Information Systems Engineering*, 57(2), 129–133. [\[CrossRef\]](#)
- Hoban, P. R., & Bucklin, R. E. (2015). Effects of internet display advertising in the purchase funnel. *Journal of Marketing Research*, 52(3), 375–393. [\[CrossRef\]](#)
- Järvinen, J., & Taiminen, H. (2016). Harnessing marketing automation for B2B content marketing. *Industrial Marketing Management*, 54, 164–175. [\[CrossRef\]](#)

- Jerath, K., Ma, L., & Park, Y. H. (2014). Consumer click behavior at a search engine. *Journal of Marketing Research*, 51(4), 480–486. [CrossRef]
- Johnston, L. (2020). *Cart abandonment email benchmarks* (Klaviyo Research Report). Klaviyo.
- Kohavi, R., Longbotham, R., Sommerfield, D., & Henne, R. M. (2013). Controlled experiments on the web. *Data Mining and Knowledge Discovery*, 18(1), 140–181. [CrossRef]
- Krosnick, J. A., & Fabrigar, L. R. (1997). Designing rating scales for effective measurement in surveys. In L. Lyberg (Ed.), *Survey measurement and process quality* (pp. 141–164). Wiley.
- Kumar, V., Bhaskaran, V., Mirchandani, R., & Shah, M. (2021). Creating a measurable social media marketing strategy: Increasing the value and ROI of intangibles and tangibles for Hokey Pokey. *Marketing Science*, 32(2), 194–212. [CrossRef]
- Kumar, V., & Reinartz, W. (2016). Creating enduring customer value. *Journal of Marketing*, 80(6), 36–68. [CrossRef]
- Kunz, W. H., & Wirtz, J. (2024). Corporate digital responsibility (CDR) in the age of AI: Implications for interactive marketing. *Journal of Research in Interactive Marketing*, 18(1), 31–37. [CrossRef]
- Lambrecht, A., & Tucker, C. (2013). When does retargeting work? *Journal of Marketing Research*, 50(5), 561–576. [CrossRef]
- Leung, F. F., Gu, F. F., Li, Y., Zhang, J. Z., & Palmatier, R. W. (2022). Influencer marketing effectiveness. *Journal of Marketing*, 86(6), 93–115. [CrossRef]
- Li, H., & Kannan, P. K. (2014). Attributing conversions in a multichannel online marketing environment. *Journal of Marketing Research*, 51(1), 40–56. [CrossRef]
- Nabout, N. A., Skiera, B., Stepanchuk, T., & Gerstmeier, E. (2012). An analysis of the profitability of fee-based compensation plans for search engine marketing. *International Journal of Research in Marketing*, 29(1), 68–80. [CrossRef]
- Nielsen, J. (1994). *Usability engineering*. Academic Press.
- Nunnally, J. C. (1978). *Psychometric theory* (2nd ed.). McGraw-Hill.
- Pavlou, P. A., & Fygenson, M. (2006). Understanding and predicting electronic commerce adoption. *MIS Quarterly*, 30(1), 115–143. [CrossRef]
- Pulizzi, J. (2012). The rise of storytelling as the new marketing. *Publishing Research Quarterly*, 28(2), 116–123. [CrossRef]
- Reichheld, F. F., & Sasser, W. E. (1990). Zero defections: Quality comes to services. *Harvard Business Review*, 68(5), 105–111. [PubMed]
- Saunders, M., Lewis, P., & Thornhill, A. (2019). *Research methods for business students* (8th ed.). Pearson.
- Spearman, C. (1904). The proof and measurement of association between two things. *American Journal of Psychology*, 15(1), 72–101. [CrossRef] [PubMed]
- Statista. (2024). *Global e-commerce market revenue 2023–2027*. Statista Research Department. Available online: <https://www.statista.com> (accessed on 1 January 2024).
- Tam, K. Y., & Ho, S. Y. (2005). Web personalization as a persuasion strategy. *Information Systems Research*, 16(3), 271–291. [CrossRef]
- Thomke, S. H. (2020). *Experimentation works: The surprising power of business experiments*. Harvard Business Review Press.
- Varian, H. R. (2007). Position auctions. *International Journal of Industrial Organization*, 25(6), 1163–1178. [CrossRef]
- Vo, D.-T., Mai, N. Q., Nguyen, L. T., Thuan, N. H., Dang-Pham, D., & Hoang, A.-P. (2024). Examining authenticity on digital touchpoint: A thematic and bibliometric review of 15 years' literature. *Journal of Research in Interactive Marketing*, 18(3), 463–484. [CrossRef]
- Wang, C. L. (2021). New frontiers and future directions in interactive marketing: Inaugural Editorial. *Journal of Research in Interactive Marketing*, 15(1), 1–9. [CrossRef]
- Wang, C. L. (2024). Editorial—What is an interactive marketing perspective and what are emerging research areas? *Journal of Research in Interactive Marketing*, 18(2), 161–165. [CrossRef]
- Wang, C. L. (2025). Editorial: Demonstrating contributions through storytelling. *Journal of Research in Interactive Marketing*, 19(1), 1–4. [CrossRef]
- Wang, C. L. (2026). Editorial: The changing landscape of marketing research in the AI era: Prospects and challenges. *Journal of Research in Interactive Marketing*, 20(1), 1–10. [CrossRef]
- Ye, G., Hudders, L., De Jans, S., & De Veirman, M. (2021). The value of influencer marketing for business. *Journal of Advertising*, 50(2), 160–178. [CrossRef]
- Zajonc, R. B. (1968). Attitudinal effects of mere exposure. *Journal of Personality and Social Psychology*, 9(2), 1–27. [CrossRef]
- Zhou, L., Dai, L., & Zhang, D. (2009). Online shopping acceptance model. *Journal of Electronic Commerce Research*, 8(1), 41–62.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.